# Geologic Period Poster

U02 / S02.3

NAME _____ DATE _____ PERIOD _____

GROUP MEMBERS _____

## THE JOB

Your group picks one geologic period and makes a poster about it. The menu on page 2 lists every period you can choose from, with the main points of each, so you can compare before you decide. The poster has to be in color.

## WHAT GOES ON THE POSTER

1. **Title.** The name of your period, large enough to read from across the room. Add when it started and ended, in millions of years ago (mya).
2. **Key Facts zone.** This holds two things. **Climate:** was it warm or cold, wet or dry, and was there ice? **Continents:** a colored map showing where the continents were during your period, each one labeled.
3. **Organisms.** Pick at least 3 important living things that evolved or first appeared in your period (the first dinosaurs, the first flowering plants). Write what each one was and why it matters.
4. **Pictures.** One picture of each organism you list, labeled with its name. Draw it or print it.
5. **Color.** The whole poster is in color, not pencil or pen alone.

## POSTER LAYOUT

**PERIOD NAME**

start and end dates, in mya

**KEY FACTS**

**CLIMATE**
warm or cold, wet or dry, ice or no ice

**CONTINENTS MAP**
colored, each continent labeled

**ORGANISMS**

At least 3. Each one gets a picture, a name, and why it matters. Add frames if you have more.

**ORGANISM 1**
picture, name, why it matters

**ORGANISM 2**
picture, name, why it matters

**ORGANISM 3**
picture, name, why it matters

Solid teal border: Key Facts. Dashed black border: Organisms. Where the zones sit on your poster is up to your group.

CHOOSE YOUR PERIOD

# The Menu: 12 Periods

mya = million years ago. Oldest at the top, youngest at the bottom. Use this to pick, then get the full facts from your class notes.

| PERIOD | CLIMATE | NEW LIFE | CONTINENTS |
|---|---|---|---|
| **PALEOZOIC ERA · 539 TO 252 MYA** | | | |
| **Cambrian**<br>539 to 485 mya | Warm. Shallow seas flooded the edges of the continents. | The Cambrian explosion: first animals with hard shells and eyes. Trilobites, Anomalocaris. All life is in the ocean. | Gondwana, a big continent mostly in the south. Smaller continents scattered near the equator. |
| **Ordovician**<br>485 to 444 mya<br>ENDS: MASS EXTINCTION | Warm with very high sea level, then an ice age near the end. | First land plants (small, mossy). Big coral reefs. Fish with no jaws. *Ohio's state fossil, the trilobite Isotelus, is from this period.* | Gondwana over the South Pole. Ohio sat under a warm shallow sea. |
| **Silurian**<br>444 to 419 mya | Warm. Ice melted and seas rose again. | First plants with stems that carry water (Cooksonia). First fish with jaws. Sea scorpions. | Two northern continents, Laurentia and Baltica, collide and close an ocean between them. |
| **Devonian**<br>419 to 359 mya<br>ENDS: MASS EXTINCTION | Mostly warm. Cooling toward the end. | The Age of Fishes: armored Dunkleosteus, early sharks. First forests. First four-legged animals; Tiktaalik, a fish with leg-like fins. *Dunkleosteus fossils come from the Cleveland Shale.* | North America and Europe joined into one northern continent. Gondwana still in the south. |
| **Carboniferous**<br>359 to 299 mya | Warm and wet with huge swamps. Ice sheets in the far south late in the period. | Swamp forests that became coal. First reptiles. Dragonfly-like insects with wings about 2 ft across. *Eastern Ohio's coal formed in these swamps.* | Gondwana and the northern continent collide. Pangaea starts to form. |
| **Permian**<br>299 to 252 mya<br>ENDS: MASS EXTINCTION | Hot and dry. Deserts in the middle of the land. | Dimetrodon and other mammal relatives. Conifer trees. Beetles. Ends with the worst mass extinction ever: about 90% of ocean species gone. | One supercontinent: Pangaea. |
| **MESOZOIC ERA · 252 TO 66 MYA** | | | |
| **Triassic**<br>252 to 201 mya<br>ENDS: MASS EXTINCTION | Hot, mostly dry. No ice at the poles. | First dinosaurs. First mammals (small). First pterosaurs. Early turtles. Ichthyosaurs in the sea. | Pangaea, still in one piece. It begins to crack apart near the end. |
| **Jurassic**<br>201 to 145 mya | Warm and humid. No ice caps. | Giant long-necked dinosaurs. Stegosaurus. First birds (Archaeopteryx). Plesiosaurs in the sea. | Pangaea splits into a northern and a southern continent. The Atlantic Ocean starts to open. |
| **Cretaceous**<br>145 to 66 mya<br>ENDS: MASS EXTINCTION | Very warm. Highest sea level of the whole era; a shallow sea cut North America in two. | First flowering plants. Tyrannosaurus rex and Triceratops. An asteroid hit about 66 mya and ended the non-bird dinosaurs. | Continents keep splitting. South America and Africa separate. India drifts north. |
| **CENOZOIC ERA · 66 MYA TO TODAY** | | | |
| **Paleogene**<br>66 to 23 mya | Hot early on, then cooling. Ice starts to build on Antarctica. | Mammals take over the empty spaces left by the dinosaurs. First whales, first horses, early primates. | Looks closer to today. India crashes into Asia and the Himalaya begin to rise. |
| **Neogene**<br>23 to 2.6 mya | Cooling and drying. Large Antarctic ice sheets. | Grasslands spread, and so do grazing animals. Mastodons. Early human ancestors (Australopithecus). | Almost the modern map. North and South America join at Panama about 3 mya. |
| **Quaternary**<br>2.6 mya to today | Repeating ice ages. Sea level dropped each time ice grew. *Ice sheets covered much of Ohio.* | Woolly mammoths, saber-toothed cats, and the first Homo sapiens (about 300,000 years ago). Many large mammals died out about 11,000 years ago. | Same positions as today. |

# Group Plan and Rubric

## OUR GROUP PLAN

| | |
|---|---|
| **Our period** | |
| **Organism 1** | |
| **Organism 2** | |
| **Organism 3** | |
| **Who makes the continents map** | |
| **Who writes the climate section** | |

## RUBRIC · 20 POINTS

| PART | FULL CREDIT | PARTIAL | LITTLE OR NONE | SCORE |
|---|---|---|---|---|
| **Title**<br>2 points | **2.** Period name and start and end dates, readable across the room. | **1.** Name only, dates missing, or hard to read. | **0.** No title. | |
| **Climate**<br>3 points | **3.** Correct. Says warm or cold, wet or dry, and gives at least one specific detail such as ice, deserts, swamps, or sea level. | **1 to 2.** One word like "hot" with no detail, or partly incorrect. | **0.** Missing or wrong. | |
| **Organisms**<br>4 points | **4.** At least 3 organisms that really first appeared in this period. Each one explained: what it was and why it matters. | **2 to 3.** Fewer than 3, or one is from a different period, or the explanations are missing. | **0 to 1.** One organism or none. | |
| **Pictures**<br>4 points | **4.** Every organism listed has a picture, and each picture is labeled with its name. | **2 to 3.** Most have pictures, or pictures are not labeled. | **0 to 1.** Few pictures or none. | |
| **Continents map**<br>4 points | **4.** Map shows where the continents were in this period. Each continent is labeled. | **2 to 3.** Map is there but unlabeled, or only partly matches the period. | **0 to 1.** No map, or just today's map. | |
| **Color and layout**<br>3 points | **3.** The whole poster is in color. Key Facts and Organisms are easy to find. Text is readable from a few feet away. | **1 to 2.** Color in some parts only, or crowded and hard to read. | **0.** Mostly black and white. | |
| | | | **TOTAL** | **/ 20** |